

Eigenvalues and Eigenvectors (Spectral Theory)

Motivation

Question: Can we efficiently calculate A^{1000} if $A = \begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix}$?

Consider the matrix $P = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$ and note that $P^{-1} = \frac{1}{7} \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$.

Let us compute $P^{-1}AP$.

$$P^{-1}AP = \frac{1}{7} \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix} = \frac{1}{7} \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 4 & 5 \\ 2 & -15 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & -5 \end{bmatrix} = D \text{ (diagonal matrix)}$$

Question: How can the above calculation help us solve the original question?

$$\begin{aligned} P^{-1}AP &= D \\ P(P^{-1}AP)P^{-1} &= PDP^{-1} \\ (PP^{-1})A(P^{-1}P) &= PDP^{-1} \\ A &= PDP^{-1} \end{aligned}$$

and therefore

$$\begin{aligned} A^{1000} &= (PDP^{-1})(PDP^{-1}) \dots (PDP^{-1})(PDP^{-1}) \\ &= PD(\bar{P}^{-1}P)D(\bar{P}^{-1}P) \dots (\bar{P}^{-1}P)D\bar{P}^{-1} \\ &= PD^{1000}\bar{P}^{-1} \end{aligned}$$

$$D^{1000} = \begin{bmatrix} 2 & 0 \\ 0 & -5 \end{bmatrix}^{1000} = \begin{bmatrix} 2^{1000} & 0 \\ 0 & (-5)^{1000} \end{bmatrix}$$

$$\begin{aligned} A^{1000} &= \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 2^{1000} & 0 \\ 0 & (-5)^{1000} \end{bmatrix} \frac{1}{7} \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} = \frac{1}{7} \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 3(2)^{1000} & 2^{1000} \\ -(5)^{1000} & 2(5)^{1000} \end{bmatrix} \\ &= \frac{1}{7} \begin{bmatrix} 3(2)^{1001} + 5^{1000} & 2^{1001} - 2(5)^{1000} \\ 3(2)^{1000} - 3(5)^{1000} & 2^{1000} + 6(5)^{1000} \end{bmatrix} \end{aligned}$$

Theorem

If A is an $n \times n$ matrix and P is an invertible $n \times n$ matrix such that $A = PDP^{-1}$, then $A^k = PD^kP^{-1}$ for each $k \in \mathbb{N}$

The process of finding an invertible matrix P and a diagonal matrix D such that $A = PDP^{-1}$ is known as diagonalizing the matrix A , and P is referred to as the diagonalizing matrix.

Questions

- Are all matrices diagonalizable? If not, when is a matrix diagonalizable?
- How can we diagonalize a matrix A ? (How can we find P and D ?)

Eigenvalues and Eigenvectors

Definition

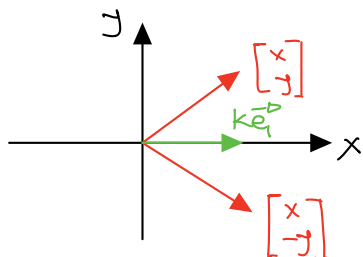
Let A be an $n \times n$ matrix, λ a real number, and $\vec{x} \neq \vec{0} \in \mathbb{R}^n$. If $A\vec{x} = \lambda\vec{x}$, then λ is an **eigenvalue** of A , and $\vec{x}$ is an **eigenvector** of A corresponding to λ .

Example: $A = \begin{bmatrix} 2 & 3 \\ 2 & 3 \end{bmatrix}$ and $\vec{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$$A\vec{x} = \begin{bmatrix} 2 & 3 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix} = 5 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 5\vec{x}$$

This means 5 is an eigenvalue of A and $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is an eigenvector of A corresponding to eigenvalue 5.

Example: Reflection wrt the x-axis in $\mathbb{R}^2$



$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = (1) \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad \vec{x} = \vec{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ is an eigenvector of A with eigenvalue $\lambda = 1$.

The Characteristic Polynomial

Question: How can we calculate the eigenvalues of a matrix?

$$A\vec{x} = \lambda\vec{x}$$

$$\lambda\vec{x} - A\vec{x} = \vec{0}$$

$$\lambda I\vec{x} - A\vec{x} = \vec{0}$$

$$(\lambda I - A)\vec{x} = \vec{0}$$

This is an SLE with coefficient matrix $\lambda I - A$ and if there exists a nontrivial solution $\vec{x}$ then $(\lambda I - A)$ is not invertible and therefore $\det(\lambda I - A) = 0$.

Example: If $A = \begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix}$ then $\lambda I - A = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix} = \begin{bmatrix} \lambda-1 & -2 \\ -3 & \lambda+4 \end{bmatrix}$

$$\det(\lambda I - A) = (\lambda-1)(\lambda+4) - 6 = 0 \rightarrow \lambda^2 + 3\lambda - 10 = 0 \rightarrow (\lambda+5)(\lambda-2) = 0$$

$$\begin{array}{l} \lambda_1 = -5 \\ \lambda_2 = 2 \end{array}$$

Definition

The **characteristic polynomial** of an $n \times n$ matrix A is defined to be $c_A(\lambda) = \det(\lambda I - A)$, where I is the identity matrix.

Question: How can we calculate the eigenvectors of a matrix?

Example: $(\lambda_1 I - A) \vec{x} = \vec{0}$

$$(2I - A) \vec{x}_2 = \vec{0}$$

$$\left[\begin{array}{cc|c} 1 & -2 & 0 \\ -3 & 6 & 0 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & -2 & 0 \\ 0 & 0 & 0 \end{array} \right] \rightarrow \vec{x}_2 = \begin{bmatrix} 2t \\ t \end{bmatrix} = t \begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad t \in \mathbb{R} \text{ and } t \neq 0$$

$$(-5I - A) \vec{x}_1 = \vec{0}$$

$$\left[\begin{array}{cc|c} -6 & -2 & 0 \\ -3 & -1 & 0 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 3 & 1 & 0 \\ 0 & 0 & 0 \end{array} \right] \rightarrow \vec{x}_1 = \begin{bmatrix} -t \\ 3t \end{bmatrix} = t \begin{bmatrix} -1 \\ 3 \end{bmatrix} \quad t \in \mathbb{R} \text{ and } t \neq 0$$

$\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 3 \end{bmatrix}$ are basic eigenvectors of A corresponding to eigenvalues (2) and (-5) respectively.